Data Management

08 NoSQL & Data Spaces

Lucas Iacono. PhD

Graz University of Technology, Know Center Research, Austria

Institute of Human-Centred Computing

Course calendar

- **Part 2: Tuning, APIs, NoSQL, Data Spaces**
  - (7) Mon. 01.06.2026: NoSQL and Data Spaces **[Last Lecture of the Course]**.
  - Tue. 02.06.2026: Deadline Exercise 3.
  - Mon. 08.06.2026: no lecture (time to prepare for VO exam #3)
  - Mon. 15.06.2026: VO exam #3 **(means no lecture)**
  - **Semester Break**

Last Lecture

- **Database APIs (Java and Python)**
  - C/C++: ODBC
  - Java: JDBC
  - Python: DB-API (psycopg2 library)
- **Object-relational mapping frameworks (ORMs)**
  - Impedance-mismatch problem
  - Map object-oriented objects to database tables
  - E.g., in Python: SQLAlchemy
- **Please also submit Exercise 3 by tomorrow! (02.06.2026)**

- **Exam (15.6): balanced distribution between i13&P2**

NoSQL Databases

HCC
Institute of
HUMAN-CENTRED COMPUTING

NoSQL Databases

What is NoSQL?

- Stands for **Not only SQL.**
- Class of **non-relational data** storage systems
- **Do not require a fixed table schema**
- **Do not** (in most cases) **use the concept of joins**
- Relaxation for one or more of the ACID properties

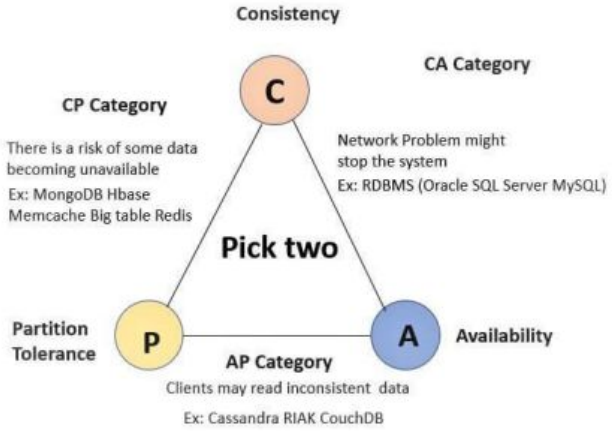
  → Atomicity, Consistency, Isolation, Durability

- Based on the CAP theorem

CAP Theorem

- Describes the **trade-offs** in the design of distributed system
- **Consistency (C):**
  - All nodes in a distributed system have the **same data at the same time.**
- **Availability (A):**
  - The system is **always operational and responsive.**
- **Partition tolerance (P):**
  - Nodes can be isolated or **communication between them can fail**

→but the **system remains functional**



A distributed system can provide at **most two** out of the three **guarantees**

HCC Institute of HUMAN-CENTRED COMPUTING

Need for NoSQL?

Data diversity:

- Growing data volumes in various formats create challenges for traditional databases

Scalability Hurdles

- Traditional databases pose limitations in scaling, often requiring costly hardware upgrades

Agile Development Challenges

- The evolving nature of software development demands a database solution that can swiftly adapt to modifications without hindering development cycles.

https://www.mongodb.com/resources/basics/databases/nosql-explained

Types of NoSQL databases

- **Key-Value** based
  - Simpler type of database where each item contains keys and values
    - **Redis**
      - An open-source, in-memory data structure store that can be used as a database, cache, and message broker
    - **Riak KV**
      - A distributed NoSQL key-value database that is highly available, fault-tolerant, and scalable
    - **Cassandra**
      - A distributed NoSQL database that provides high availability and scalability without compromising performance







Types of NoSQL databases

- **Column based**
  - Store data in columns instead of rows, optimized for analytical processing
    - **HBase**
      - An open-source, column-oriented database management system that runs on top of Hadoop Distributed File System
    - **Hypertable**
      - An open-source, high-performance, distributed database modeled after Google's Bigtable
    - **Cassandra***
      - While columns are an essential data modeling element, they are stored based on a partition key, making it more of a hybrid between a column-oriented and KV store







Types of NoSQL databases

- **Document based**
  - Store **JSON-like data in documents**, each comprising key-value pairs with **values** encompassing **strings, numbers, booleans, arrays, or objects**
    - **MongoDB**
      - A general-purpose, document-based database that emphasizes flexibility and developer productivity



    - **Couchbase**
      - A distributed, document-based database that provides high performance, scalability, and availability



    - **Apache Solr**
      - Document schema-based search engine that allows users to index and search large volumes of semi-structured and unstructured data



HCC
Institute of HUMAN-CENTRED COMPUTING

Types of NoSQL databases

- **Graph based**
  - **Store data using nodes** for e.g., people, places, and things, with **edges capturing information about relationships** between nodes
    - **Neo4j**
      - Designed to handle data with complex relationships and interconnections
    - **OrientDB**
      - A multi-model database, supporting graph but also document, key-value and object-oriented paradigms
    - **ArangoDB**
      - Similar to OrientDB without the object-oriented paradigm





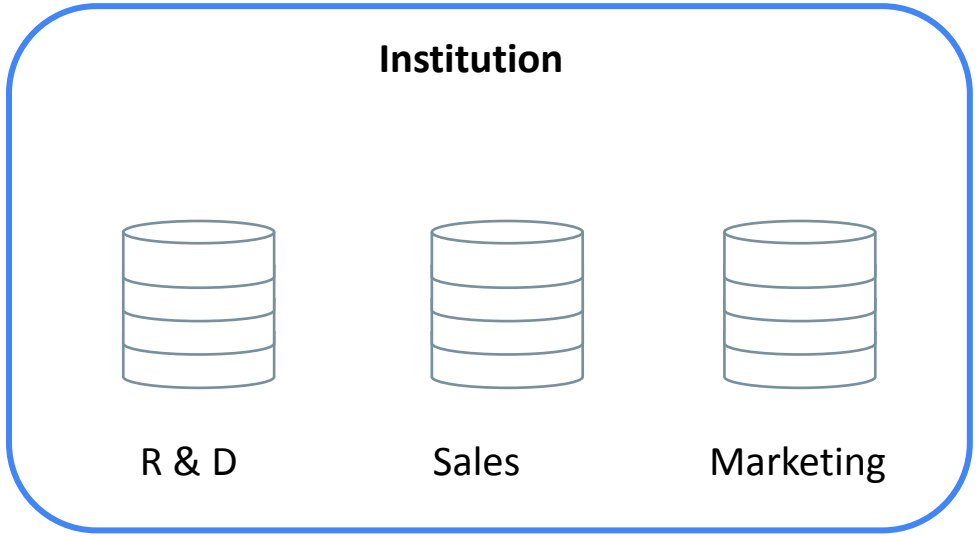


Evolution of Data Sharing

At the beginning...

Data Silos (Intra-department Data Sharing)

- Information isolated, stored in local databases (or repositories) according to the structure of the company.
- Duplication of storage.
- Increase of economic costs.
- Difficulties to share data across departments.
- Heterogeneous technologies.
- Data inconsistency.

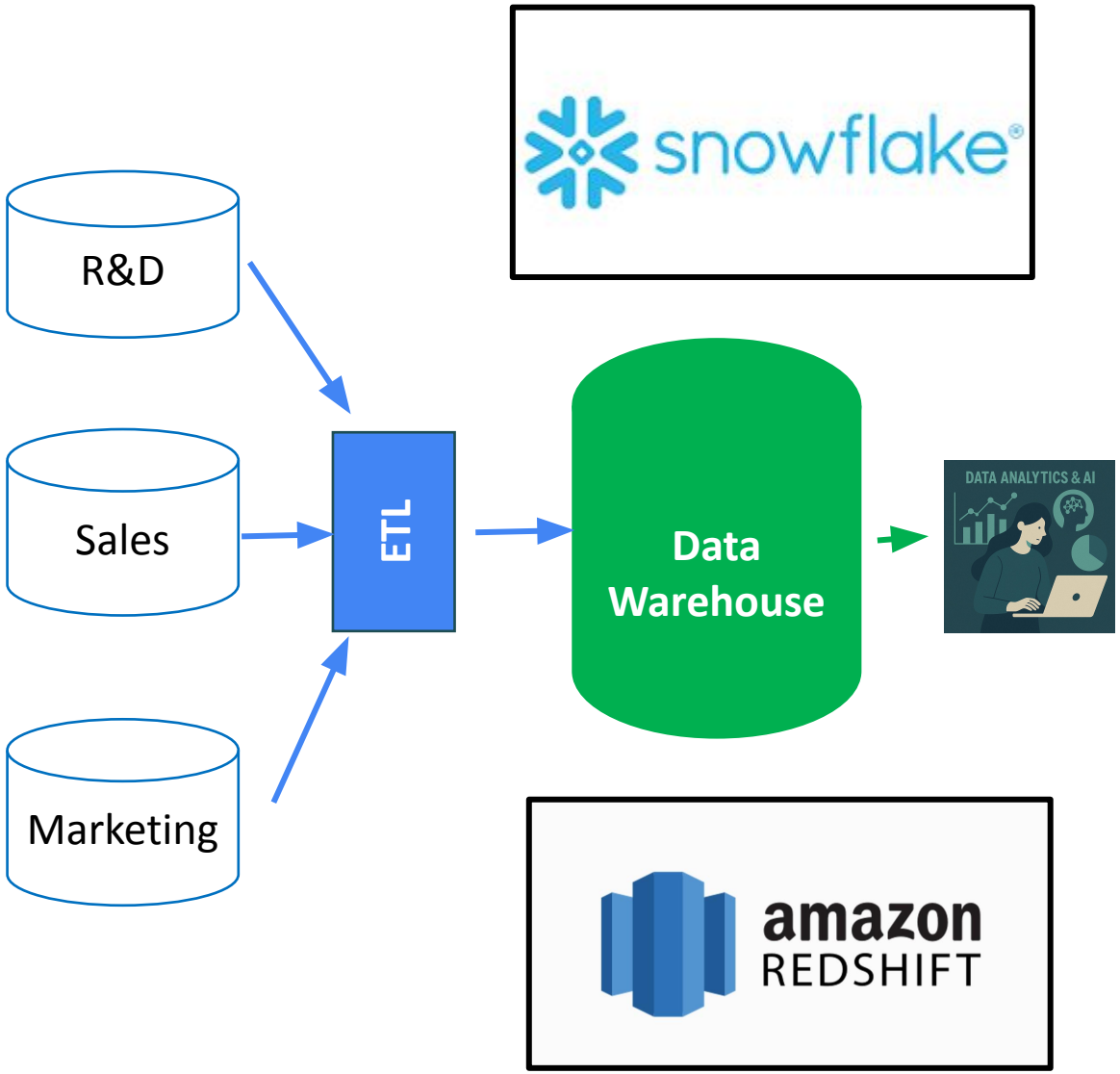


Let's break the silos...

Data warehouse (Intra-organization data sharing)

Definition: is a **centralized** data management architecture designed to support business intelligence (BI) by enabling **analytical queries** over **integrated, cleaned, and structured data** from multiple, often heterogeneous, internal sources.

- **Optimized for analytical workloads:** Data warehouses prioritize **read performance,** enabling **fast complex queries over large datasets**, while sacrificing write speed.
- **Data consolidation:** Integrates data from various internal systems through **ETL** processes (extraction from data sources, cleaning, normalization, loading) **-> consistency and quality.**

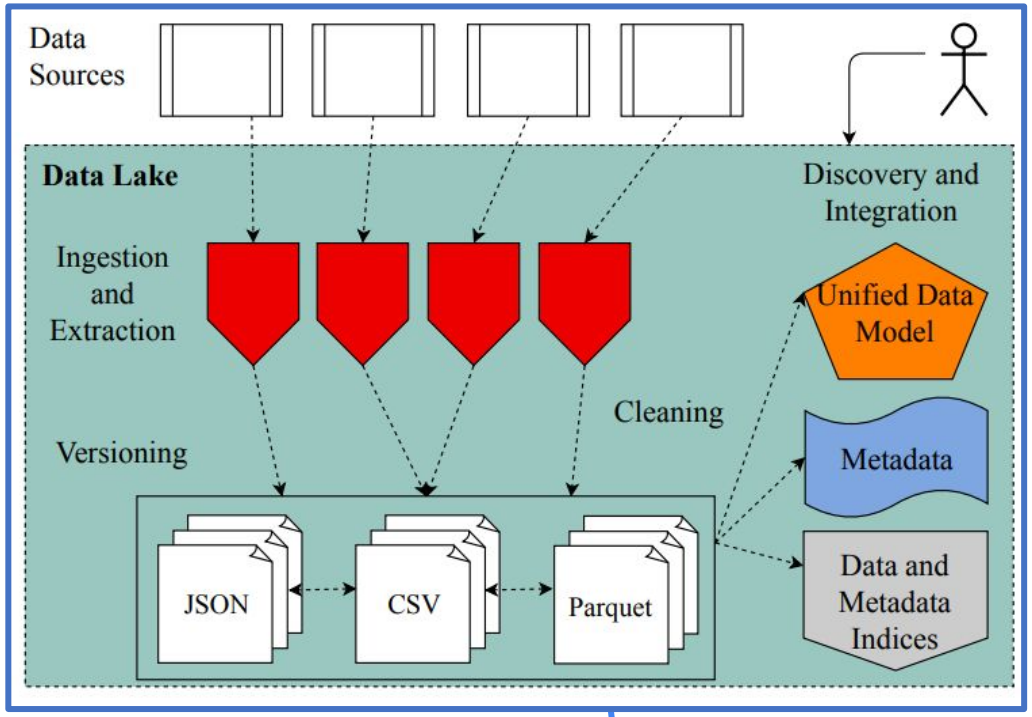


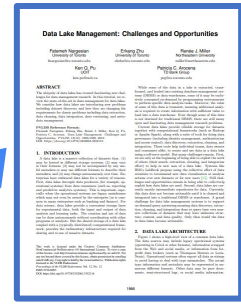
HCC Institute of HUMAN-CENTRED COMPUTING

Let's distribute and add some chaos

Data Lake (Intra/Inter organization data sharing)

- **Definition:** It's a massive collection of datasets that may…
  - be **hosted** in different storage systems
  - **vary** in their formats
  - **not** be accompanied by any useful **metadata** or may use different formats to describe their metadata
  - **change** autonomously **over time.**





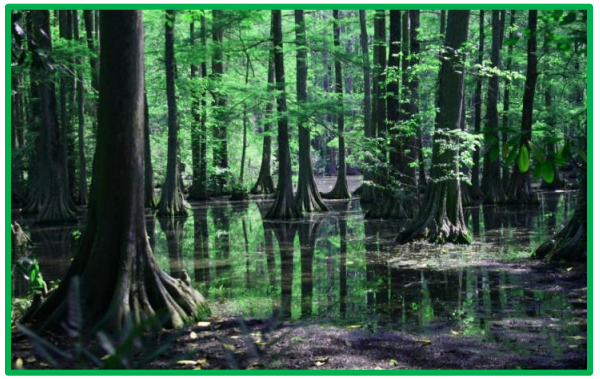
[*] Nargesian, et.al. (2019). Data lake management: challenges and opportunities. *Proceedings of the VLDB Endowment*

Let's distribute and add some chaos

Data Lake (Inter/Intra organization data sharing)

- **Store Everything:**
  - Data added as-is (Not modified in place.)
  - Store **all kinds of data,** no matter its structure.
- **No Pre-Planning Required:**
  - **No** need for defining a **fixed schema** (data structure) before adding data.
  - Useful for situations where **analysis to perform is not yet clear.**
- **File-Based Storage:**
  - Data stored as **raw files**, open formats (**CSV, JSON,** etc.).
  - Files can serve as in**puts or intermediate outputs** for further **processing**.
- **Scalable and Agile:**
  - Data lakes rely on distributed storage to handle huge datasets.

Data Swamp!!!!



Let’s add some order again

Data Lakehouse (Intra-organizational data sharing)





- **Definition:**
  - A data lakehouse is a new, open data management architecture that **combines** the flexibility, cost-efficiency, and scale of **data lakes** with the **data management advantages of data warehouses**, enabling business intelligence (BI) and machine learning (ML) on all data. (Databricks)

Key aspects

- **Metadata** layers for data lakes
- New query engine designs providing **high-performance SQL** execution on data lakes
- **Optimized access** for data science and machine learning tools.

Is everything fix after data lakehouses?

- Break silos
- Support data science and ML
- Enable scalability
- Support structured and non-structured data
- Provide data governance
- Solve all the challenges of data sharing inter- organizations



Why???

Is everything fix after data lakehouses?

- **Organizational silos persist**
  - **Each company** has its **own lakehouse** → difficult to share data **across entities.**
- **Data usage control**
  - Once **data is shared, control is lost.**
- **Lack of trust in the ecosystem**
  - How can you **trust** that a partner will follow the rules?
  - How can you **ensure** regulatory compliance? **(e.g. GDPR)**
- **Fragmented governance**
  - Each **organization** implements **its own rules**.
  - There are **no** common **standards or interoperability** mechanisms.

What is a Data Space

The data connector

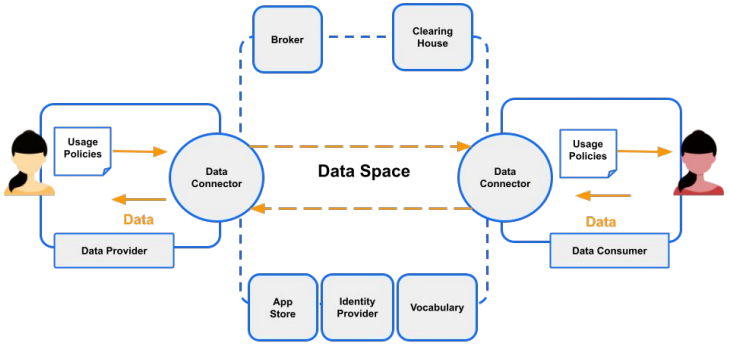
- Let's the AI to explain:



What is a Data Space?

In one shot:

- A data space is a **secure and standardized digital infrastructure** that enables **trusted data exchange** and **data-based services** among various stakeholders.



A formal definition:

- According to IDSA*: is a **virtual space** that provides a **standardized framework** for **data exchange**, based on **common protocols and formats**, as well as **secure and trusted data sharing mechanisms**.

*International Data Space Association

What is a Data Space?

Data spaces pillars:

- **Data sovereignty:** data owners retain **control over their data** and can determine **who** can use it and **under what conditions** (data usage policies and digital contracts).
- **Standardization:** **protocols** designed to enable **data sharing between organizations** from different countries and regions i**ndependently of the underlaying IT infrastructure.**
- **Federated storage and decentralization:** data are stored in different places and **only meta-data is shared** on the data space.
- **Governance:** Data remains **under the control of the data creator**

HCC
Institute of
HUMAN-CENTRED COMPUTING

Data Spaces Technologies

TU Graz

HCC
Institute of
HUMAN-CENTRED COMPUTING

Data Space Protocol

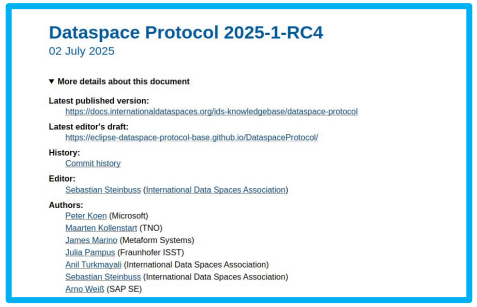
Specifications designed to facilitate **interoperable data sharing** between entities governed by usage control and **based on Web technologies** (DCAT catalogs, ODRL, HTTPS, etc).

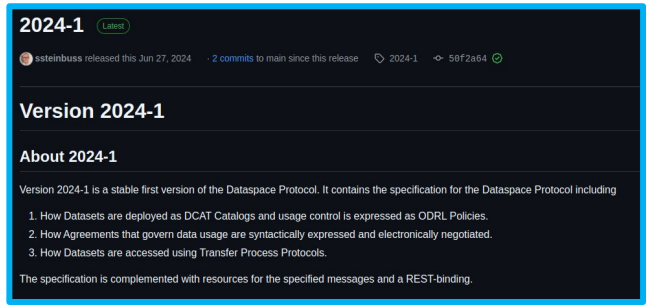
These specifications define the schemas and protocols required for entities to **publish** data, **negotiate** Agreements, and **access data** in a data space

Versions:

- **Previous:** 0.8 and 2024-1 IDSA
- **Current:** Eclipse Data Space Protocol Project

- More info: https://docs.internationaldataspaces.org/ids-knowledgebase/dataspace-protocol





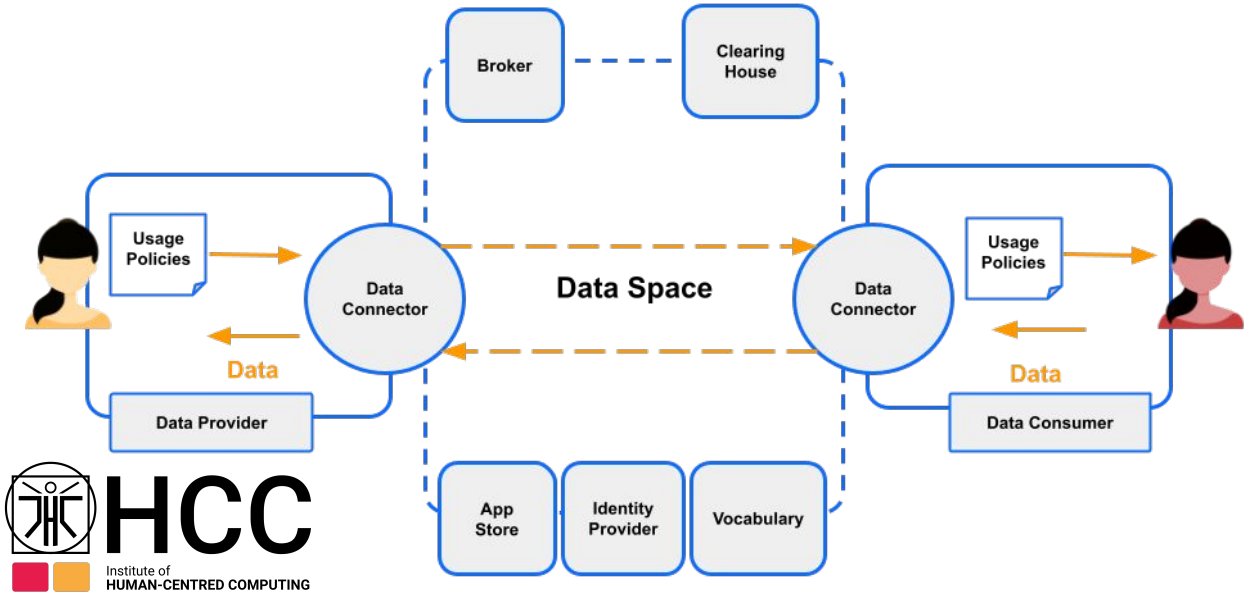
Data Space Protocol

Term	Definition
Agreement	Policy associated with a specific Dataset signed by both the Provider and Consumer. It's a result of a Contract Negotiation and is associated with exactly one Dataset.
Catalog	Collection of entries representing Datasets offered by a Provider Participant.
Connector (Data Service)	Participant Agent that produces Agreements and manages Dataset sharing.
Consumer	Participant Agent that requests access to an offered Dataset.
Provider	Participant Agent that offers a Dataset.
Contract Negotiation	Set of interactions between a Provider and Consumer that establish an Agreement. It is an instantiation of the **state machine** of a **Contract Negotiation Protocol.**
Contract Negotiation Protocol	Set of allowable Message Type sequences defined as a state machine.
Credential Issuer	System that issues verifiable credentials for a Participant and Participant Agents.
Dataset	Data or a technical service that can be shared by a Participant.
Dataspace	Set of technical services that facilitate interoperable Dataset sharing between entities.
Dataspace Authority	Entity that manages a Dataspace. The form and capabilities of a Dataspace Authority are not covered in these specifications.

The data connector

- The **data connector** is a **software component** that **enables participants to attach usage policies to their data in a data space**, it **enforces the usage policies and track data provenance**.

- The data connector act as a **gateway** for services apps and software



SOVITY

FIWARE

ECLIPSE DATASPACE

nexyo

HCC
Institute of HUMAN-CENTRED COMPUTING

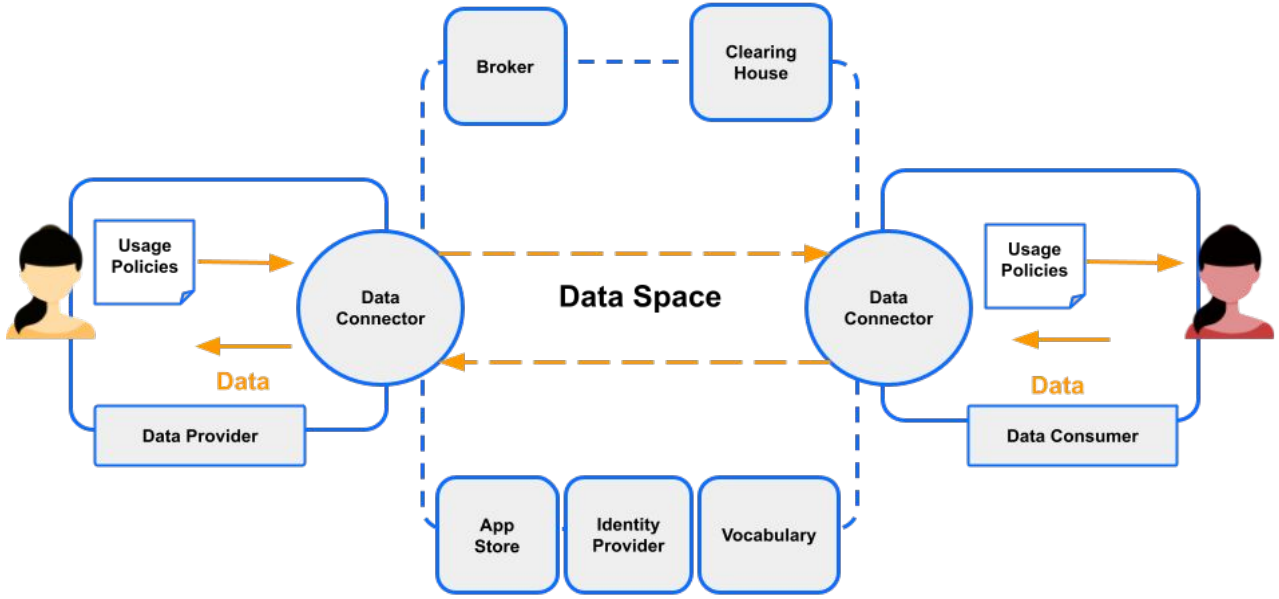
Minimum Viable Data Space

Let's do our data space:

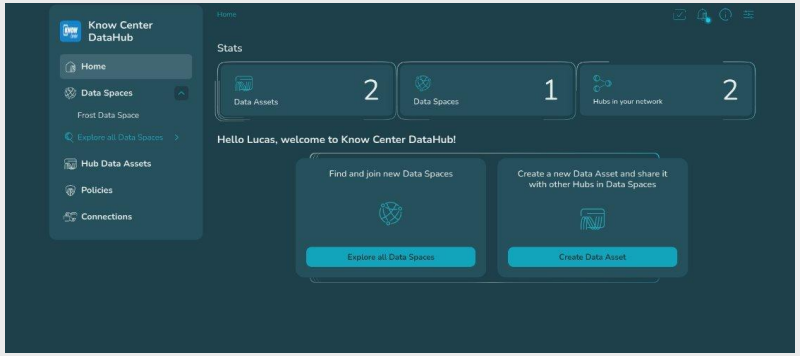
Minimum viable data space [*] is a combination of components that enable the creation of a data space with **just enough features** to be usable for secure and sovereign data exchange.

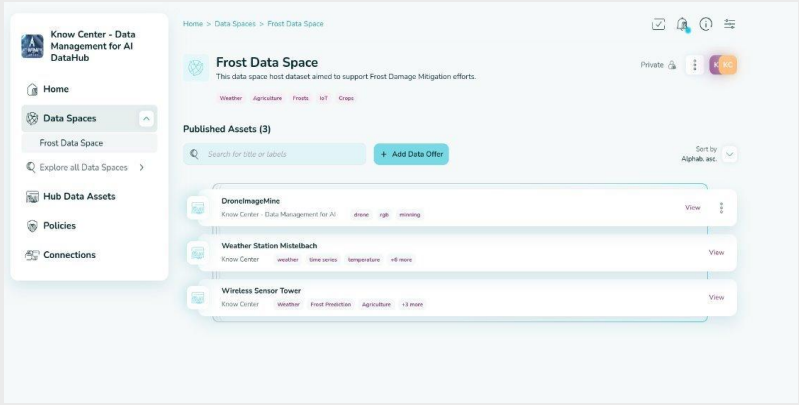
Main Components

- Two connectors.
- An identity provider.
- Optional and additional components:
  - Metadata broker
  - App store
  - Clearing house
  - Vocabulary provider

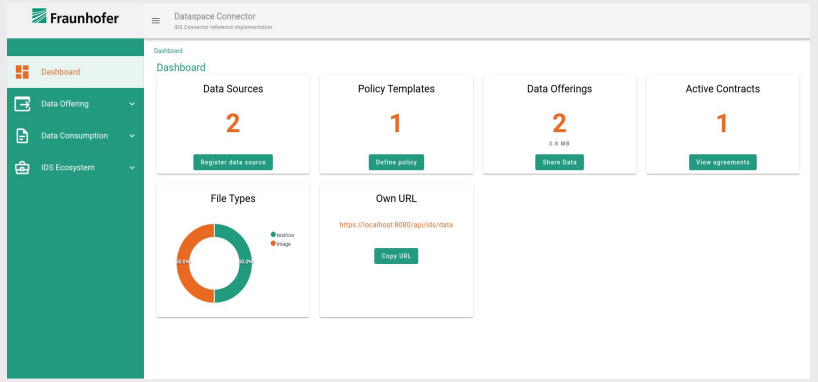


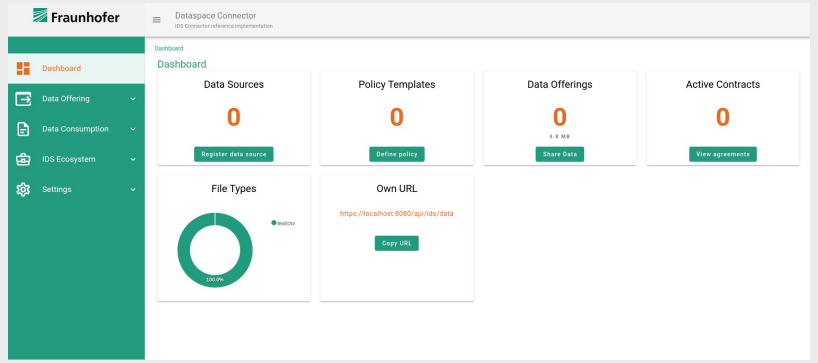
[*] Definition by the International Data Spaces Association.
https://docs.internationaldataspaces.org/ids-knowledgebase/v/ids-reference-testbed/minimum-viable-data-space/mvds



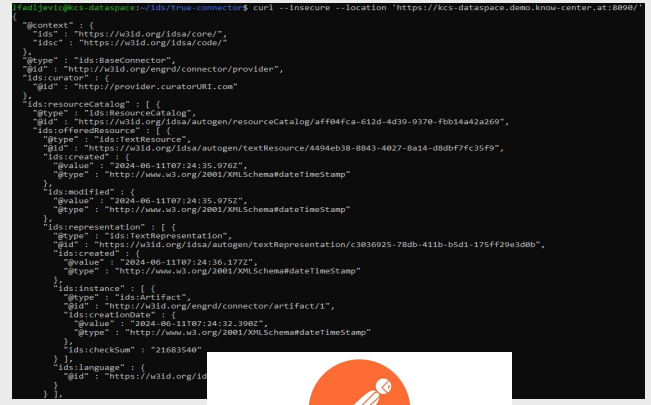


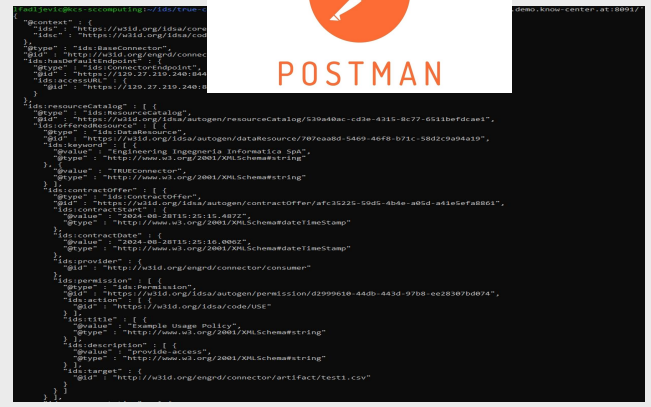
nexyo





Fraunhofer





POSTMAN

TRUE Connector

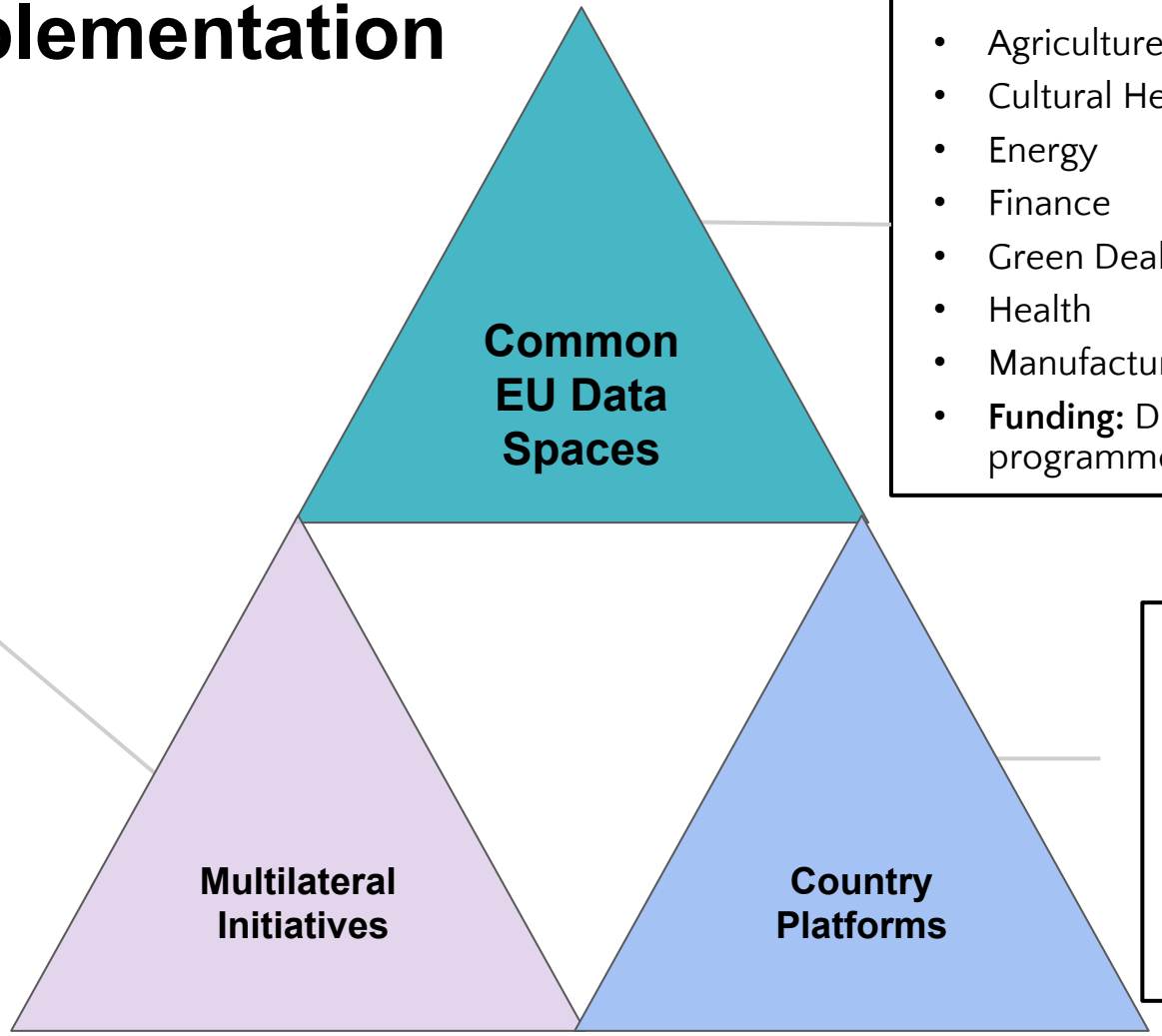
HCC
Institute of
HUMAN-CENTRED COMPUTING

MVDS – **Usecase implemented on 3 different tech stacks**

Feature	Nexyo	Fraunhofer	True
IDSA Certfication	NA	Yes	Yes
Connector Core	EDC	EDC	TRUE
Licence	NA	Open source (Apache 2.0)	Open source (AGPL version 3) 1
GUI	Yes (Nexyo Data Hub)	Yes (EDC Dashboard)	No
Hardware requirements	Cloud-based (Azure)	Dockers	CPU: 4 core (8 threads) Memory: at least 2GB Docker Containers
Software Requirements	Cloud-based (Azure)	Java 17 + Docker	Linux + Docker
Data Sources / Sinks	Amazon S3, MS Sharepoint, Google Drive, Azure Blob, Google Cloud	Data bases	NA
Policies	Pre-defined and ad-hoc templates	Ad-hoc templates	NA
Maintained	Yes	No	Yes
API	Yes (WEB)	Yes	Yes (REST)
Identity Management	Basic auth (Basic access authorization, providing username and password)	Centralized/DAPS, IDS certificates (X.509v3)	Centralized (X.509)
Communication Protocol	IDS Multipart, IDS Communication protocol (IDSCP)	IDS Multipart, IDS Communication protocol (IDSCP)	IDS Multipart, IDS protocol (IDSCP)

Data Space Implementations

Data Spaces Implementation



Common EU Data Spaces (14):

- Agriculture
- Cultural Heritage
- Energy
- Finance
- Green Deal
- Health
- Manufacturing
- **Funding:** Digital Europe / Horizon Europe programmes

- **Domain specific data space initiatives for industries**
- Gaia-X
- Catena-X
- Manufacturing-X
- energy dat-X.
- Funding: EU & national funding.

National & regional data space initiatives: (federated across Europe)

- Tourism Data Spaces in Spain and Italy.
- Datahub.tirol.
- Digital Product Passport

Source; https://digital-strategy.ec.europa.eu/en/library/staff-working-document-data-spaces & https://gaia-x.eu

Example: Catena-X Data Space for Automotive



Research Challenges in Data Spaces

TU Graz

HCC
Institute of
HUMAN-CENTRED COMPUTING

Some Research Challenges in Data Spaces

Connection of Data Spaces with Data Science Pipelines.

Current initiatives focus on protocols and software for data spaces but **not in the role of data spaces as data providers for** data analysis pipelines.

More efforts → Facilitate the connection of data spaces with data science and AI pipelines (e.g. capacity to manage large volumes of data, costs of cloud deployments, among others).

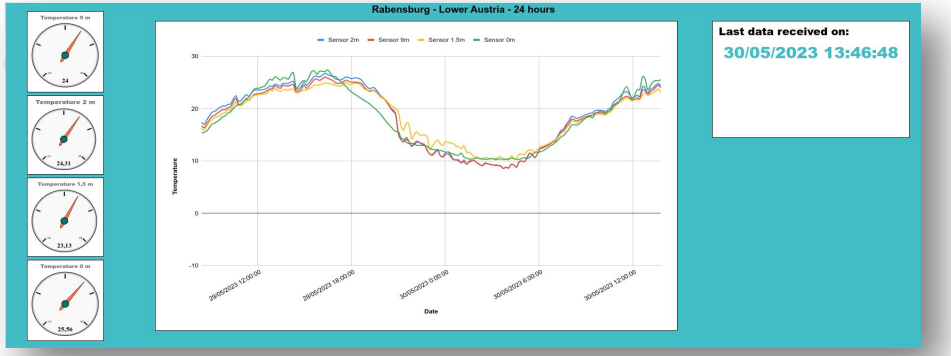
Some Research Challenges in Data Spaces

Scalability.

- This is a critical challenge in data spaces due to **growing data**, requiring **cloud-based solutions validated through performance testing.**
- Development and validation of scalable cloud solutions for data spaces, assessing different services based on criteria like compute time, cost, reliability, and availability.

KnowFROSTs: A Data Space for Frost Damage Mitigation





HCC
Institute of
HUMAN-CENTRED COMPUTING

Motivation


Kronen Zeitung
STEIERMARK

STEIRER-BAUERN KÄMPFEN

Mit Eis, Wind und Feuer gegen verheerenden Frost


steiermark ORF.at

LANDWIRTSCHAFT

Frost: Obstbauern befürchten Ernteausfälle

Bis Freitag macht der Frühling eine Pause, und vor allem kalte Nächte stehen in der Steiermark bevor. Sinken die Temperaturen zu sehr in den Minusbereich, könnten Obstkulturen Schaden nehmen. Schutzmaßnahmen seien aufwendig und teuer, klagen Obstbauern.


ORF

„Thema“ am 24. April: Kampf dem Frost – Bauern bangen um ihre Existenz

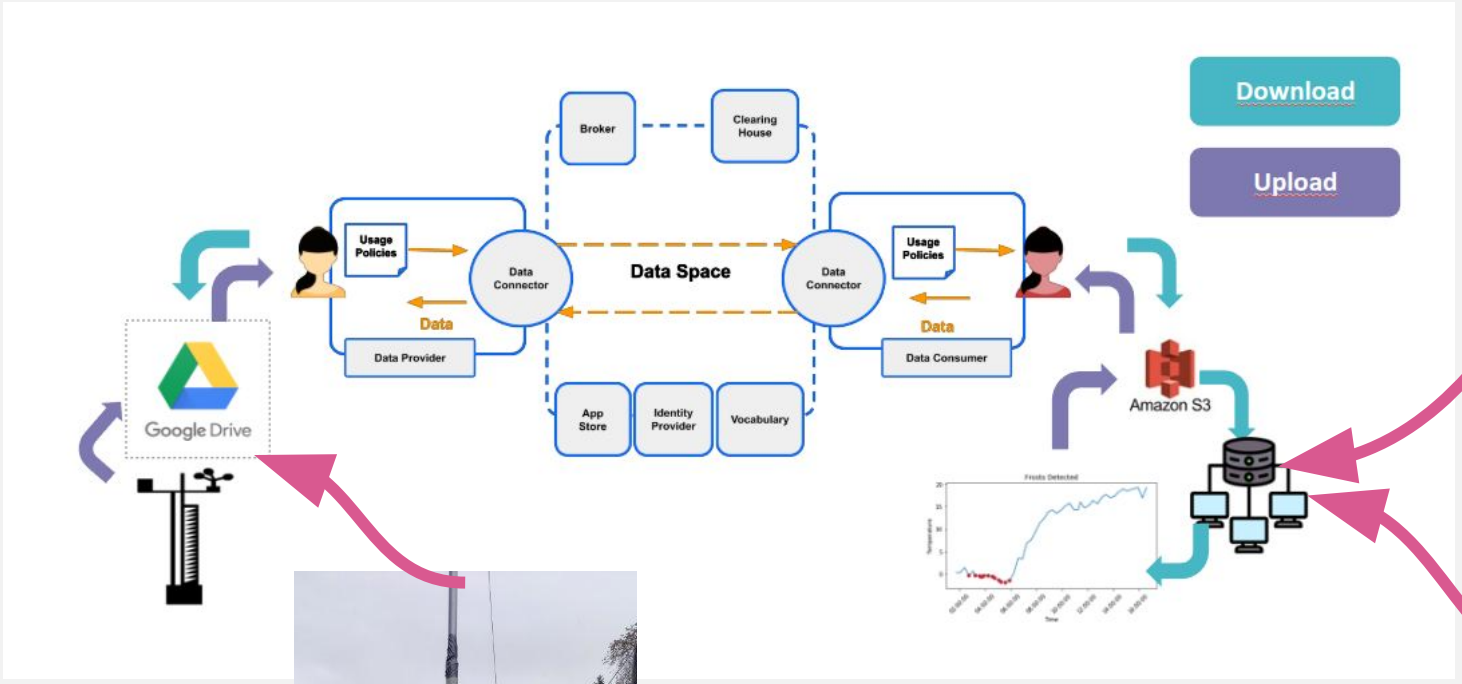
In 2016, frost damage to fruit trees and vines was reported by Styrian farmers at around **200 million euros**. This is just one example of the many crop failures caused by frost.

At the Know Center, we develop technological solutions that can help farmers minimize their frost damage by 90%.

Our focus is on:

- AI & Numerical weather prediction models
- IoT
- **Data Spaces**

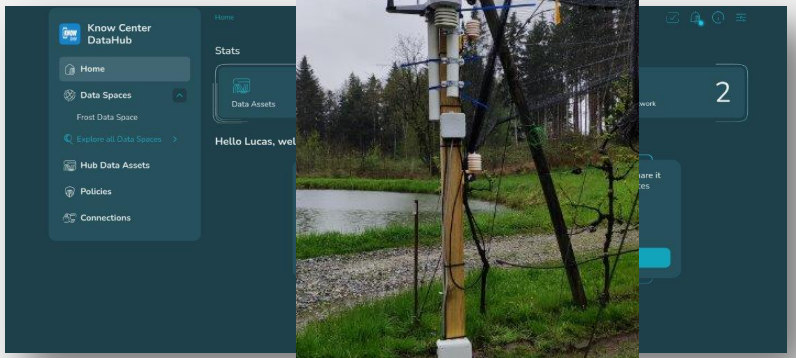
HCC
Institute of HUMAN-CENTRED COMPUTING

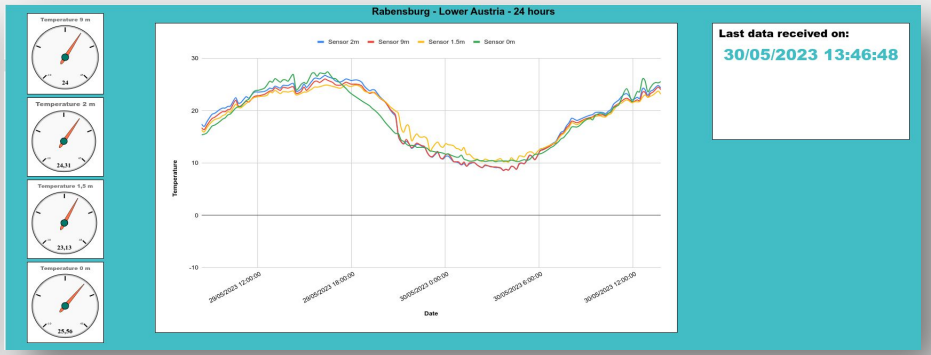


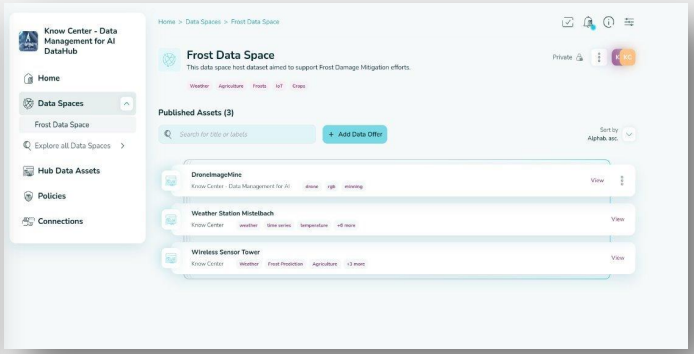
Neural Prophet

- **Objective:** Predict 2-meter temperature.
- **Training:** Based on cleaned historical data with min-level resolution.
- **Applications:** Prediction of extreme events (e.g., frost).
- Decision support for agriculture management.





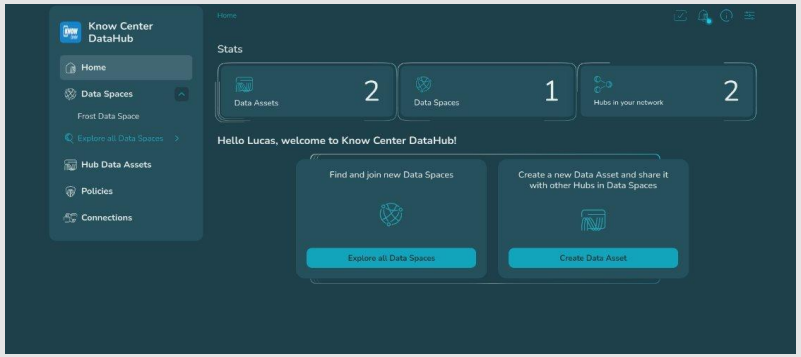


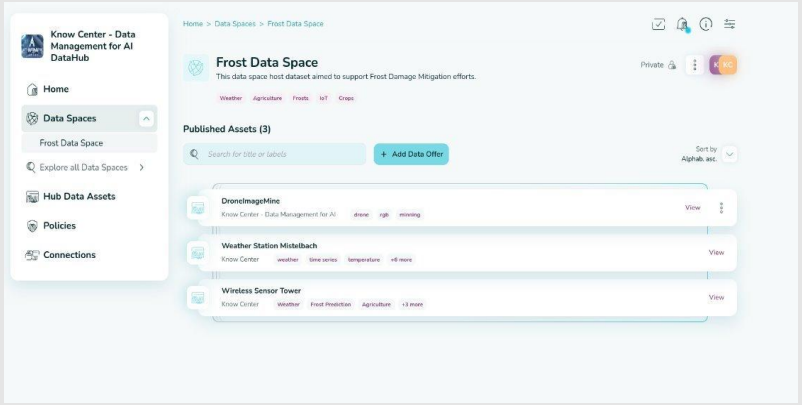


Click here to access live data!!

Access Data Space (live):
Data Provider
Data Consumer

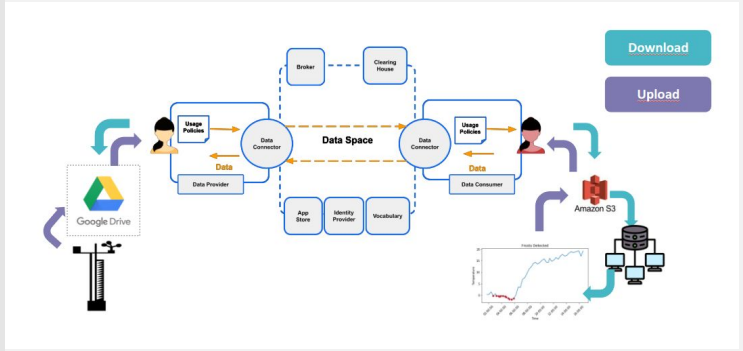
KnowFROSTs

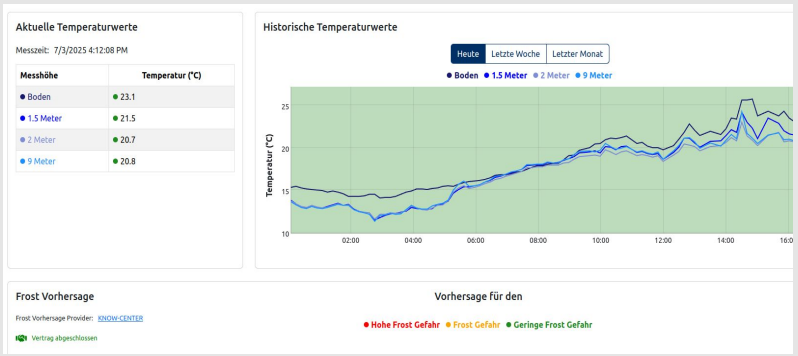


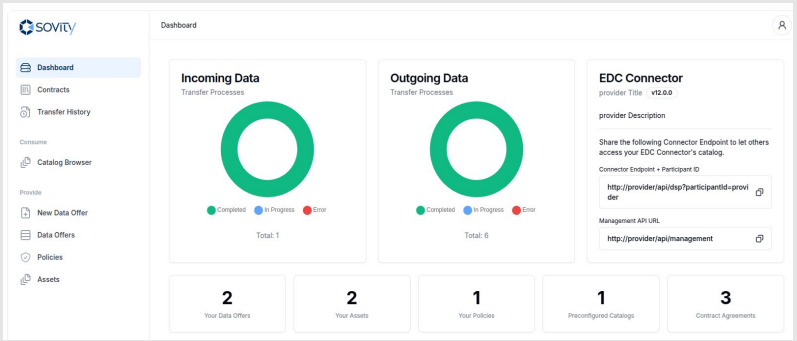


nexyo

Demonstrator available
Data Sink







SOVITY

Provider
Consumer

Two different tech stacks

KnowFROSTs

Course calendar

- **Part 2: Tuning, APIs, NoSQL, Data Spaces**
  - Tue. 02.06.2026: Deadline Exercise 3.
  - Mon. 08.06.2026: no lecture (time to prepare for VO exam #3)
  - Mon. 15.06.2026: VO exam #3 **(means no lecture)**
  - **Semester Break**



HCC
Institute of
HUMAN-CENTRED COMPUTING